Depth-First Search Continued: Coloring and Testing for Bipartiteness

Design and Analysis of Algorithms
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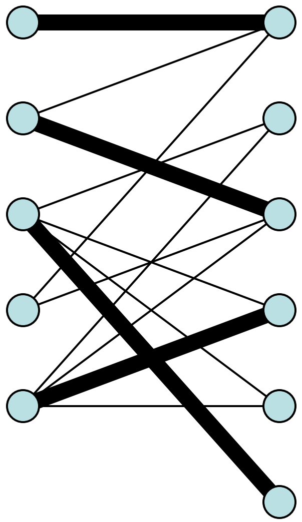
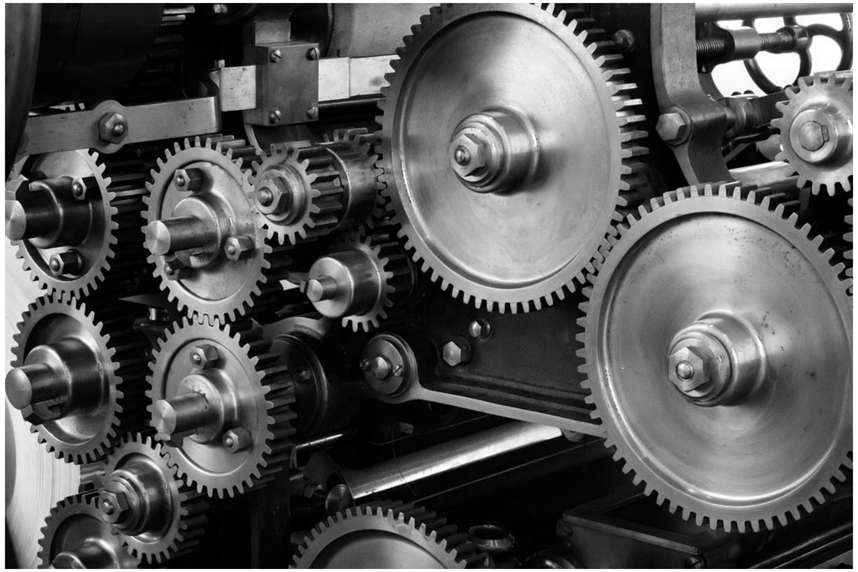
Bipartite Graphs

- Very prominent class of graphs; e.g., for matching problems.





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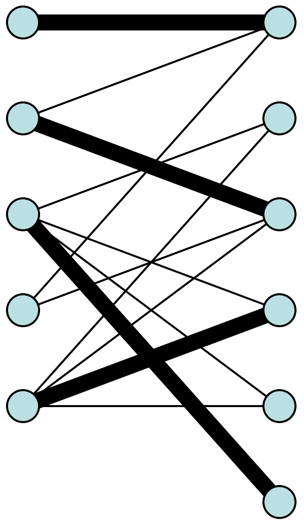
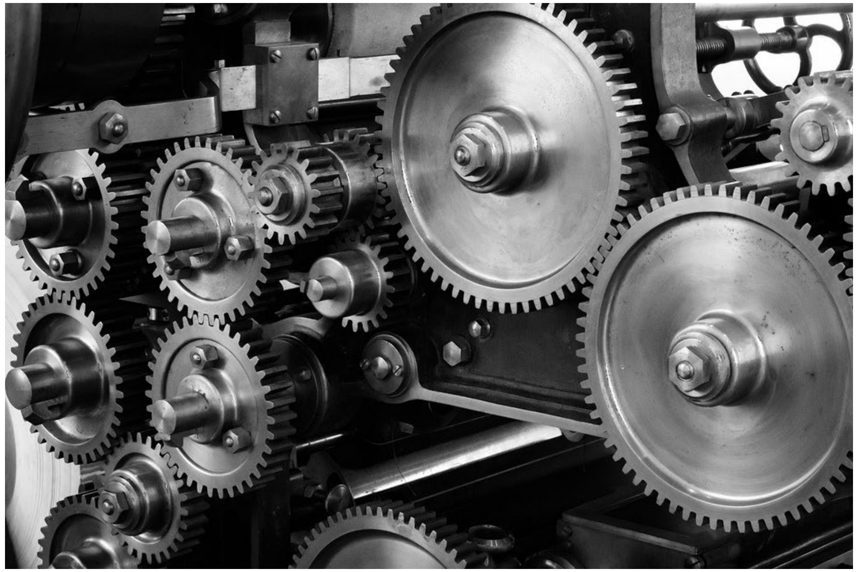




- Is it easy to test if a graph is bipartite?

  YES – just keep track of "parity" during a full DFS.

2SAT

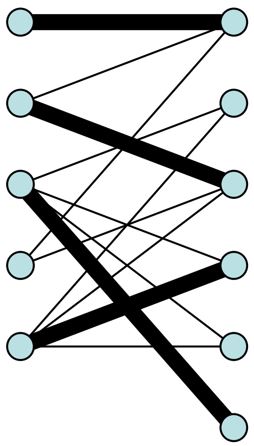
- Given a Boolean expression with 2-variable clauses all “AND”ed together:

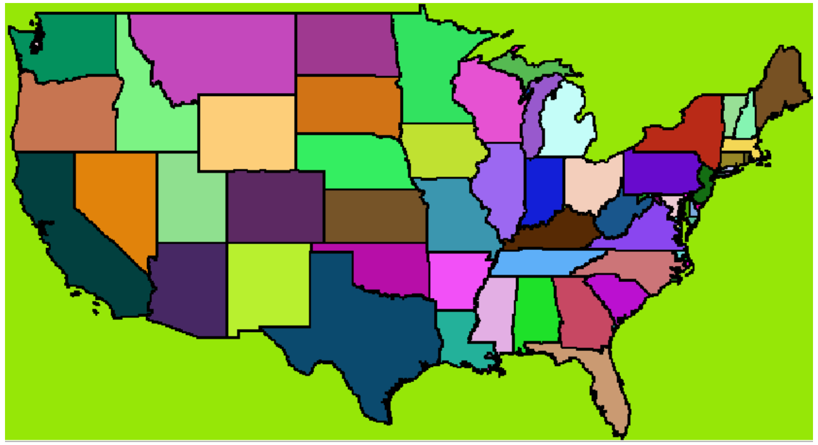
(x OR y) AND (not x OR z) AND (z OR y) AND (not y OR z)

- Can we assign values to the variables to make the entire expression true? (i.e., can we make every clause true?)
- A similar linear-time algorithm can solve this problem.
- Note: 3SAT is a famous NP-hard problem!

Coloring and Bipartite Graphs

- Bipartite graphs are “2-colorable”





- Famous result: planar graphs are 4-colorable
- Each color class is an “independent set” (a set of nodes no two of which are connected)
- Unfortunately, NP-hard to find the “chromatic number” of a general graph (min # of colors needed to color it)